

The What, Why and How of Testing



This article is for developers and those who are interested in testing. It covers the need for testing, as well as the basic concepts and the various techniques used in the process. It also explores some open source tools for testing.

Here's a quick quiz to see how many of us understand what testing is all about and why we do it—mentally note 'Yes/True' or 'No/False' regarding each of these statements:

- Testing is the same as quality control.
- We do testing to ensure that the product is 100 per cent defect-free.
- Testing is very easy, but a boring job.
- Testing does not involve any creativity.

Needless to say, most would respond with a 'Yes' to all the above questions. However, that would be incorrect; and I will explain why, soon. Many (in fact, most) of us tend to feel testing is a low-grade and boring task. But it is not so—it is perhaps the most challenging task anybody can take up. No development activity will teach you as much as testing will, provided it's done correctly, and with

the right attitude. This is my personal opinion based on my experience so far; other people may have different opinions, though.

We know that a *defect* is an error in a program that prevents it from producing the expected results. Is this definition complete? We will learn more about that as we proceed. For now, here is a preliminary understanding: Testing is done to show that there are no errors in the program; and that the program is performing its intended functions correctly—i.e., doing what it was designed to do.

However, while that understanding is correct, it is not complete. What about when a program does something it is not supposed to do? So, the point is, do not test to show that your program contains no defects. Instead, *assume that your program contains defects, and then start testing to dig them out.*